Calculus 1 formula sheet

The formulas of Calculus 1 (Mathematical Analysis 1) with their hypotheses: elementary functions and trigonometry, complex numbers, standard limits, derivatives and Taylor expansions, integrals and improper integrals, series and differential equations.

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Sets, induction and the binomial theorem

The proofs and the examples are in the chapter Sets, induction and elementary functions.

Supremum and infimum with epsilon

$$L = \sup A \iff \begin{cases} a \le L \quad \text{for every } a \in A \\ \text{for every } \varepsilon > 0 \text{ there exists } a \in A \text{ with } a > L - \varepsilon \end{cases}$$

$$\ell = \inf A \iff \begin{cases} a \ge \ell \quad \text{for every } a \in A \\ \text{for every } \varepsilon > 0 \text{ there exists } a \in A \text{ with } a < \ell + \varepsilon \end{cases}$$

It holds for $A \subseteq \mathbb{R}$ nonempty and $L, \ell \in \mathbb{R}$. If A is not bounded above one sets $\sup A = +\infty$, and this is equivalent to saying that for every $M > 0$ there exists $a \in A$ with $a > M$. The maximum of A exists *if and only if* $\sup A \in A$, and in that case it coincides with $\sup A$.

Principle of induction

$$\begin{cases} P(n_0) \text{ is true} \\ P(n) \implies P(n+1) \quad \text{for every } n \ge n_0 \end{cases} \implies \quad P(n) \text{ is true for every } n \ge n_0$$

The first condition is the base case, the second the inductive step, in which $P(n)$ is the inductive hypothesis. The conclusion requires that *both* conditions be satisfied.

Standard sums

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6},$$

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2, \qquad \sum_{k=0}^{n} q^k = \frac{1-q^{n+1}}{1-q} \quad (q \ne 1)$$

The first three hold for every $n \ge 1$, the geometric sum for every $n \in \mathbb{N}$ and every $q \ne 1$; for $q = 1$ the sum equals $n+1$. They are proved by induction on n.

Bernoulli's inequality

$$(1+x)^n \geq 1+nx$$

It holds for every $x \geq -1$ and every $n \in \mathbb{N}$. With $x = 1$ one obtains $2^n \geq 1+n$.

Factorial and binomial coefficient

$$n! = n \cdot (n-1) \cdots 2 \cdot 1, \qquad 0! = 1, \qquad (n+1)! = (n+1) \cdot n!,$$

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}$$

The binomial coefficient is defined for $n, k \in \mathbb{N}$ with $0 \leq k \leq n$ and counts the subsets with k elements of a set with n elements. In computations the factorials are simplified with the recursive formula, for example $\dfrac{10!}{8!} = 10 \cdot 9 = 90$.

Properties of the binomial coefficients

$$\binom{n}{0} = \binom{n}{n} = 1, \qquad \binom{n}{k} = \binom{n}{n-k},$$

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1} \quad (0 \leq k \leq n-1),$$

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n, \qquad \sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0 \quad (n \geq 1)$$

The second line is Pascal's rule, which generates Pascal's triangle. The two sums in the third line follow from the binomial theorem with $a = b = 1$ and with $a = 1, b = -1$.

Binomial theorem

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k = a^n + n\,a^{n-1}b + \binom{n}{2} a^{n-2}b^2 + \cdots + b^n,$$

$$(a+b)^2 = a^2 + 2ab + b^2, \qquad (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

It holds for every $a, b \in \mathbb{R}$ and every $n \in \mathbb{N}$. The term of the expansion that contains b^k is $\binom{n}{k} a^{n-k} b^k$.

Elementary functions and natural domain

The properties of the elementary functions and the rules for the domain are in the chapter Sets, induction and elementary functions.

Properties of powers

$$a^x\,a^y = a^{x+y}, \qquad \frac{a^x}{a^y} = a^{x-y}, \qquad (a^x)^y = a^{xy},$$

$$(ab)^x = a^x\,b^x, \qquad a^{-x} = \frac{1}{a^x}, \qquad a^0 = 1, \qquad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

They hold for $a, b > 0$ and $x, y \in \mathbb{R}$; in the last one $m \in \mathbb{Z}$ and $n \geq 1$. With integer exponents the same rules also hold for negative bases; a zero base requires positive exponents.

Roots and absolute value

$$\sqrt{x^2} = |x|, \qquad \left(\sqrt{x}\right)^2 = x \;\; (x \geq 0), \qquad \sqrt{a\,b} = \sqrt{a}\,\sqrt{b} \;\; (a, b \geq 0),$$

$$\sqrt[n]{x^n} = |x| \;\; (n \text{ even}), \qquad \sqrt[n]{x^n} = x \;\; (n \text{ odd})$$

The first holds for every $x \in \mathbb{R}$; for $x = -3$ one has $\sqrt{9} = 3 = |-3|$. Roots with odd index are defined on all of $\mathbb{R}$, those with even index on $[0, +\infty)$.

Absolute value and inequalities

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \qquad |xy| = |x|\,|y|, \qquad |x + y| \leq |x| + |y|$$

$$|f(x)| < a \iff -a < f(x) < a,$$

$$|f(x)| > a \iff f(x) < -a \;\text{ or }\; f(x) > a$$

The two equivalences hold for $a > 0$. For $a < 0$ the first inequality has no solutions and the second is satisfied at every point of the domain of f.

Logarithm and its properties

$$y = \log_a x \iff a^y = x, \qquad a^{\log_a x} = x, \qquad \log_a a^x = x,$$

$$\log_a(xy) = \log_a x + \log_a y, \qquad \log_a \frac{x}{y} = \log_a x - \log_a y,$$

$$\log_a x^\alpha = \alpha \log_a x, \qquad \log_a 1 = 0, \qquad \log_a a = 1$$

They hold for $a > 0$, $a \neq 1$, $x, y > 0$ and $\alpha \in \mathbb{R}$; in the identity $\log_a a^x = x$ the variable x is any real number. The logarithm to base e is denoted by $\ln x$.

Change of base and exponential to base e

$$\log_a x = \frac{\log_b x}{\log_b a} = \frac{\ln x}{\ln a}, \qquad a^x = e^{x \ln a}, \qquad f(x)^{g(x)} = e^{g(x) \ln f(x)}$$

The change of base holds for $a, b > 0$, $a \neq 1$, $b \neq 1$ and $x > 0$. Writing $f(x)^{g(x)}$ as an exponential requires $f(x) > 0$ and is used in limits and in derivatives of powers with variable base and exponent.

Warning

The logarithm of a sum is different from the sum of the logarithms. For example $\ln(1+1) = \ln 2$, whereas $\ln 1 + \ln 1 = 0$. Moreover the equality $\log_a x^2 = 2 \log_a x$ requires $x > 0$; for $x \neq 0$ one writes $\log_a x^2 = 2 \log_a |x|$, because with $x = -2$ one has $\log_2 4 = 2$ and $\log_2(-2)$ is not defined.

Exponential equations and inequalities

$$a^x = b \iff x = \log_a b \quad (b > 0),$$

$$a^{f(x)} > a^{g(x)} \iff f(x) > g(x) \quad (a > 1),$$

$$a^{f(x)} > a^{g(x)} \iff f(x) < g(x) \quad (0 < a < 1)$$

They hold for $a > 0$, $a \neq 1$. For $b \leq 0$ the equation $a^x = b$ has no solutions, because the exponential takes only positive values. The equivalences between inequalities follow from the strict monotonicity of the exponential.

Logarithmic inequalities

$$a > 1: \quad \log_a f(x) > \log_a g(x) \iff \begin{cases} f(x) > g(x) \\ g(x) > 0 \end{cases}$$

$$0 < a < 1: \quad \log_a f(x) > \log_a g(x) \iff \begin{cases} f(x) < g(x) \\ f(x) > 0 \end{cases}$$

The system contains the existence condition of the smaller argument, and the other existence condition follows from the two in the system. With a number c on the right-hand side one writes $c = \log_a a^c$, therefore for $a > 1$ the inequality $\log_a f(x) > c$ is equivalent to $f(x) > a^c$.

Conditions for the natural domain

$\dfrac{N(x)}{D(x)}$	$D(x) \neq 0$
$\sqrt[n]{g(x)},\ n$ even	$g(x) \geq 0$
$\log_a g(x)$	$g(x) > 0$
$\tan g(x)$	$g(x) \neq \dfrac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$
$\arcsin g(x),\ \arccos g(x)$	$-1 \leq g(x) \leq 1$
$f(x)^{g(x)}$	$f(x) > 0$

The conditions relative to the different operations of an expression must hold at the same point, therefore they are collected in a system; the natural domain is the set of solutions of the system. Roots with odd index, exponentials, sin, cos and arctan impose no conditions on the argument.

Symmetries and periodicity

$$f(-x) = f(x)\ \text{(even)}, \qquad f(-x) = -f(x)\ \text{(odd)},$$

$$f(x+T) = f(x)\ \text{(periodic with period } T),$$

$$\sin(\omega x),\ \cos(\omega x)\ \text{have period}\ \frac{2\pi}{|\omega|}, \qquad \tan(\omega x)\ \text{has period}\ \frac{\pi}{|\omega|}$$

Even and odd require a domain symmetric with respect to the origin; the graph of an even function is symmetric with respect to the y-axis, that of an odd function with respect to the origin. The periods in the second line are the minimal periods and hold for $\omega \neq 0$.

Transformations of the graph

$f(x) + k$	vertical translation by k
$f(x - k)$	horizontal translation by k
$a\, f(x)$	vertical stretch by a factor a
$f(ax)$	horizontal compression by a factor a
$-f(x),\ \ f(-x)$	reflection across the x-axis, across the y-axis
$\lvert f(x)\rvert$	the part with $f(x) < 0$ is reflected across the x-axis
$f(\lvert x\rvert)$	the part with $x \geq 0$ is kept and reflected across the y-axis

The translations are upward and to the right for $k > 0$, downward and to the left for $k < 0$. Stretch and compression refer to $a > 1$; for $0 < a < 1$ the two effects are exchanged.

Trigonometric formulas

The definitions of the trigonometric functions and the main identities are in the chapter Sets, induction and elementary functions.

Standard values of sine, cosine and tangent

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\tan x$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef.	0

The angles are measured in radians, and the angle of 180° corresponds to π. The values in the other quadrants are obtained with the reduction formulas.

Pythagorean identity and tangent

$$\sin^2 x + \cos^2 x = 1, \qquad \tan x = \frac{\sin x}{\cos x}, \qquad 1 + \tan^2 x = \frac{1}{\cos^2 x}$$

The first holds for every $x \in \mathbb{R}$, the other two for $x \neq \frac{\pi}{2} + k\pi$ with $k \in \mathbb{Z}$. Sine and cosine take values in $[-1, 1]$ and have period 2π; the tangent has period π.

Reduction formulas

$$\sin(-x) = -\sin x, \qquad \cos(-x) = \cos x, \qquad \tan(-x) = -\tan x,$$

$$\sin(\pi - x) = \sin x, \qquad \cos(\pi - x) = -\cos x,$$

$$\sin(\pi + x) = -\sin x, \qquad \cos(\pi + x) = -\cos x,$$

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x, \qquad \cos\left(\frac{\pi}{2} - x\right) = \sin x$$

They hold for every $x \in \mathbb{R}$, and for the tangent at every point where it is defined. Sine and tangent are odd functions, the cosine is an even function.

Addition and subtraction formulas

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y,$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y,$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

The first two hold for every $x, y \in \mathbb{R}$, the third when $\tan x$, $\tan y$ and $\tan(x \pm y)$ are defined.

Double-angle and power-reduction formulas

$$\sin 2x = 2\sin x\cos x, \qquad \cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x,$$

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \cos^2 x = \frac{1+\cos 2x}{2}, \qquad \tan 2x = \frac{2\tan x}{1-\tan^2 x}$$

They hold for every $x \in \mathbb{R}$, the last one when $\tan x$ and $\tan 2x$ are defined. The power-reduction formulas, $\sin^2 x$ and $\cos^2 x$ expressed through $\cos 2x$, lower the degree and are used in integrals.

Half-angle substitution formulas

$$t = \tan\frac{x}{2}: \qquad \sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \tan x = \frac{2t}{1-t^2}$$

They hold for $x \neq \pi + 2k\pi$ with $k \in \mathbb{Z}$; the third also requires $t^2 \neq 1$, that is $x \neq \frac{\pi}{2} + k\pi$. They transform a rational expression in $\sin x$ and $\cos x$ into a rational expression in t.

Linear combination of sine and cosine

$$a\sin x + b\cos x = R\sin(x+\varphi), \qquad R = \sqrt{a^2+b^2}, \quad \cos\varphi = \frac{a}{R}, \quad \sin\varphi = \frac{b}{R}$$

It holds for $a, b \in \mathbb{R}$ not both zero. It reduces a linear equation in sine and cosine, such as $\sin x + \cos x = 1$, to an elementary equation; in this case $\sin x + \cos x = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$.

Elementary trigonometric equations

$$\sin x = m \iff x = \arcsin m + 2k\pi \ \text{ or } \ x = \pi - \arcsin m + 2k\pi,$$

$$\cos x = m \iff x = \pm\arccos m + 2k\pi, \qquad \tan x = m \iff x = \arctan m + k\pi$$

The first two have solutions *only* for $-1 \le m \le 1$, the third for every $m \in \mathbb{R}$; in all of them $k \in \mathbb{Z}$. Inequalities are solved on the unit circle over an interval whose length equals the period and are extended by periodicity.

Arcsine, arccosine and arctangent

$$\arcsin : [-1,1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \qquad \arccos : [-1,1] \to [0,\pi], \qquad \arctan : \mathbb{R} \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$

$$\arcsin x + \arccos x = \frac{\pi}{2}, \qquad \arctan x + \arctan\frac{1}{x} = \pm\frac{\pi}{2}$$

The first identity holds for $x \in [-1,1]$; in the second the sign is $+$ for $x > 0$ and $-$ for $x < 0$. Moreover $\sin(\arcsin x) = x$ for every $x \in [-1,1]$, whereas $\arcsin(\sin x) = x$ holds *only* for $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Hyperbolic functions

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \cosh^2 x - \sinh^2 x = 1$$

They are defined on $\mathbb{R}$. The function $\sinh x$ is odd with range $\mathbb{R}$, the function $\cosh x$ is even with range $[1, +\infty)$. The inverse of the hyperbolic sine is $\ln\left(x + \sqrt{x^2+1}\right)$.

Complex numbers

The chapter Complex numbers contains the proofs and the examples.

Operations in algebraic form

$$z = a + ib, \qquad i^2 = -1, \qquad \mathrm{Re}(z) = a, \qquad \mathrm{Im}(z) = b,$$

$$(a+ib)(c+id) = (ac - bd) + i(ad + bc),$$

$$\frac{a+ib}{c+id} = \frac{(a+ib)(c-id)}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} + i\,\frac{bc-ad}{c^2+d^2}$$

They hold for $a, b, c, d \in \mathbb{R}$, and the quotient for $c + id \neq 0$. The powers of i repeat with period 4, since $i^2 = -1$, $i^3 = -i$, $i^4 = 1$; therefore $i^n = i^r$, where r is the remainder of the division of n by 4.

Conjugate and modulus

$$\bar{z} = a - ib, \qquad |z| = \sqrt{a^2+b^2}, \qquad z\,\bar{z} = |z|^2, \qquad \frac{1}{z} = \frac{\bar{z}}{|z|^2},$$

$$z + \bar{z} = 2\,\mathrm{Re}(z), \qquad z - \bar{z} = 2i\,\mathrm{Im}(z), \qquad \overline{z+w} = \bar{z} + \bar{w}, \qquad \overline{zw} = \bar{z}\,\bar{w},$$

$$|zw| = |z|\,|w|, \qquad \left|\frac{z}{w}\right| = \frac{|z|}{|w|}, \qquad |z+w| \leq |z| + |w|$$

They hold for $z = a + ib$ and w arbitrary complex numbers, with $z \neq 0$ in the formula for the reciprocal and $w \neq 0$ in the quotient. The number z is real *if and only if* $z = \bar{z}$.

Trigonometric form and principal argument

$$z = \rho\,(\cos\theta + i\sin\theta), \qquad \rho = |z|, \qquad \cos\theta = \frac{a}{\rho}, \quad \sin\theta = \frac{b}{\rho}$$

$$\theta = \begin{cases} \arctan\dfrac{b}{a} & \text{if } a > 0 \\ \arctan\dfrac{b}{a} + \pi & \text{if } a < 0,\ b \geq 0 \\ \arctan\dfrac{b}{a} - \pi & \text{if } a < 0,\ b < 0 \\ \dfrac{\pi}{2} & \text{if } a = 0,\ b > 0 \\ -\dfrac{\pi}{2} & \text{if } a = 0,\ b < 0 \end{cases}$$

It holds for $z = a + ib \neq 0$. The formula gives the principal argument $\theta \in (-\pi, \pi]$, and every other argument differs from θ by an integer multiple of 2π.

Euler's formula and exponential form

$$e^{i\theta} = \cos\theta + i\sin\theta, \qquad z = \rho\,e^{i\theta}, \qquad \bar{z} = \rho\,e^{-i\theta},$$

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

They hold for $\theta \in \mathbb{R}$ and $\rho = |z| > 0$. The exponentials $e^{i\theta}$ and $e^{i\varphi}$ coincide *if and only if* $\theta - \varphi$ is an integer multiple of 2π. Frequently used values are $e^{i\pi} = -1$ and $e^{i\frac{\pi}{2}} = i$.

Product, quotient and De Moivre's formula

$$z\,w = \rho\sigma\,e^{i(\theta+\varphi)}, \qquad \frac{z}{w} = \frac{\rho}{\sigma}\,e^{i(\theta-\varphi)},$$

$$z^n = \rho^n e^{in\theta} = \rho^n\big(\cos(n\theta) + i\sin(n\theta)\big)$$

They hold for $z = \rho e^{i\theta}$ and $w = \sigma e^{i\varphi}$ with $\rho, \sigma > 0$, and for every integer $n \geq 1$. In the product the moduli are multiplied and the arguments are added.

n-th roots

$$z^n = w \iff z = z_k = \sqrt[n]{\sigma}\;e^{\,i\frac{\varphi+2k\pi}{n}}, \qquad k = 0, 1, \ldots, n-1$$

It holds for $w = \sigma e^{i\varphi} \neq 0$ and $n \geq 1$ integer. The solutions are *exactly* n, have modulus $\sqrt[n]{\sigma}$ and are the vertices of a regular polygon with n sides centered at the origin. The n-th roots of unity are $e^{\,i\frac{2k\pi}{n}}$; for $n = 4$ they are 1, i, -1, $-i$.

Quadratic equations in the complex field

$$az^2 + bz + c = 0, \qquad \Delta = b^2 - 4ac < 0, \qquad z = \frac{-b \pm i\sqrt{-\Delta}}{2a}$$

It holds for $a, b, c \in \mathbb{R}$ with $a \neq 0$, and the two solutions are complex conjugates. With complex coefficients the solutions are $z = \dfrac{-b \pm \delta}{2a}$, where δ is either of the two complex square roots of Δ.

Equations with conjugate and modulus

$$z = x + iy, \qquad \bar{z} = x - iy, \qquad |z| = \sqrt{x^2 + y^2}, \qquad A = B \iff \begin{cases} \mathrm{Re}(A) = \mathrm{Re}(B) \\ \mathrm{Im}(A) = \mathrm{Im}(B) \end{cases}$$

It is used for equations in which $\bar{z}$, $|z|$, $\mathrm{Re}(z)$ or $\mathrm{Im}(z)$ appear. Setting $z = x + iy$ with $x, y \in \mathbb{R}$, the equality of the two sides A and B is equivalent to the system of two real equations in the unknowns x and y.

Loci in the complex plane

$$|z - z_0| = r \ \text{ circle with center } z_0 \text{ and radius } r,$$

$$|z - z_0| \leq r \ \text{ disk with center } z_0 \text{ and radius } r,$$

$$|z - z_1| = |z - z_2| \ \text{ perpendicular bisector of the segment with endpoints } z_1, z_2,$$

$$\mathrm{Re}(z) = k \ \text{ line } x = k, \qquad \mathrm{Im}(z) = k \ \text{ line } y = k$$

They hold for $r > 0$, $z_1 \neq z_2$ and $k \in \mathbb{R}$. The number $|z - w|$ is the distance between the points z and w.

Fundamental theorem of algebra and real factorization

$$p(z) = a_n(z - z_1)(z - z_2)\cdots(z - z_n),$$

$$(z - z_0)(z - \bar{z}_0) = z^2 - 2\alpha z + \alpha^2 + \beta^2 \quad (z_0 = \alpha + i\beta)$$

Every polynomial of degree $n \geq 1$ with complex coefficients has n roots $z_1, \ldots, z_n$, counted with multiplicity. If the coefficients are real, the non-real roots appear in conjugate pairs with the same multiplicity, and the product of the two conjugate factors is a real quadratic with discriminant $-4\beta^2 < 0$.

Limits, indeterminate forms and continuity

The definitions and the theorems are in the chapter Limits and continuity.

Operations with infinite limits

$$L + (\pm\infty) = \pm\infty, \qquad (+\infty) + (+\infty) = +\infty, \qquad (-\infty) + (-\infty) = -\infty,$$

$$L \cdot (\pm\infty) = \pm\infty \ \ (L > 0), \qquad L \cdot (\pm\infty) = \mp\infty \ \ (L < 0), \qquad (+\infty) \cdot (\pm\infty) = \pm\infty,$$

$$\frac{L}{\pm\infty} = 0, \qquad \frac{L}{0^+} = +\infty \ \ (L > 0), \qquad \frac{L}{0^-} = -\infty \ \ (L > 0)$$

Here $L \in \mathbb{R}$ is the limit of one of the two terms, and 0^+, 0^- denote a function that tends to 0 while staying positive or negative; for $L < 0$ the signs in the last two are exchanged. If $f \to L$ and $g \to M$ with $L, M \in \mathbb{R}$, the limits of sum, product and quotient are computed with the operations on L and M, with $M \neq 0$ in the quotient.

Indeterminate forms

$$+\infty - \infty, \qquad 0 \cdot \infty, \qquad \frac{0}{0}, \qquad \frac{\infty}{\infty}, \qquad 1^\infty, \qquad 0^0, \qquad \infty^0$$

In these cases the limit depends on the functions and not only on the limits of the parts. The last three are reduced to $0 \cdot \infty$ with $f(x)^{g(x)} = e^{\,g(x)\ln f(x)}$, valid for $f(x) > 0$. The quotient $\dfrac{L}{0}$ with $L \neq 0$ gives an infinite limit, with the sign determined by the sign of L and by that of the denominator.

Limits of the elementary functions

$$\lim_{x\to+\infty} e^x = +\infty, \qquad \lim_{x\to-\infty} e^x = 0, \qquad \lim_{x\to 0^+} \ln x = -\infty, \qquad \lim_{x\to+\infty} \ln x = +\infty,$$

$$\lim_{x\to+\infty} x^\alpha = +\infty \ \ (\alpha > 0), \qquad \lim_{x\to+\infty} x^\alpha = 0 \ \ (\alpha < 0), \qquad \lim_{x\to\pm\infty} \arctan x = \pm\frac{\pi}{2},$$

$$\lim_{x\to+\infty} a^x = +\infty \ \ (a > 1), \qquad \lim_{x\to+\infty} a^x = 0 \ \ (0 < a < 1), \qquad \lim_{x\to 0^\pm} \frac{1}{x} = \pm\infty$$

For $x \to -\infty$ one has $a^x \to 0$ if $a > 1$ and $a^x \to +\infty$ if $0 < a < 1$. Every elementary function is continuous on its domain, therefore at a point of the domain the limit is the value of the function.

Polynomials and quotients of polynomials at infinity

$$\lim_{x\to\pm\infty} \frac{a_n x^n + \cdots + a_0}{b_m x^m + \cdots + b_0} = \lim_{x\to\pm\infty} \frac{a_n x^n}{b_m x^m} = \begin{cases} \dfrac{a_n}{b_m} & \text{if } n = m \\ 0 & \text{if } n < m \\ \pm\infty & \text{if } n > m \end{cases}$$

It holds for $a_n \neq 0$ and $b_m \neq 0$, because for $x \to \pm\infty$ a polynomial is asymptotic to its term of highest degree. For $n > m$ the sign of the infinity is the sign of $\dfrac{a_n}{b_m}\, x^{\,n-m}$.

Factorizations and the factor theorem

$$a^2 - b^2 = (a-b)(a+b), \qquad a^3 \mp b^3 = (a \mp b)(a^2 \pm ab + b^2),$$

$$a^n - b^n = (a-b)\left(a^{n-1} + a^{n-2}b + \cdots + ab^{n-2} + b^{n-1}\right),$$

$$ax^2 + bx + c = a\,(x - x_1)(x - x_2)$$

The last one holds for $a \neq 0$ and $\Delta = b^2 - 4ac \geq 0$, with x_1 and x_2 roots of the quadratic, coincident if $\Delta = 0$. By the **factor theorem**, if $P(x_0) = 0$ the polynomial $P(x)$ is divisible by $x - x_0$; in the $\frac{0}{0}$ limits of quotients of polynomials for $x \to x_0$ the factor $x - x_0$ is factored out and canceled.

Rationalization

$$\sqrt{A} - \sqrt{B} = \frac{A - B}{\sqrt{A} + \sqrt{B}}, \qquad \sqrt[3]{A} - \sqrt[3]{B} = \frac{A - B}{\sqrt[3]{A^2} + \sqrt[3]{AB} + \sqrt[3]{B^2}},$$

$$\lim_{x \to +\infty} \left(\sqrt{x^2 + bx + c} - x\right) = \frac{b}{2}$$

The first holds for $A, B \geq 0$ not both zero, the second for A, B real and not both zero; the third for every $b, c \in \mathbb{R}$. They are used for the forms $+\infty - \infty$ and $\frac{0}{0}$ with roots. For $x \to -\infty$ one has $\sqrt{x^2} = |x| = -x$, therefore $\sqrt{x^2 + bx + c} \sim -x$.

Change of variable in limits

$$\lim_{x \to x_0} f(g(x)) = \lim_{y \to y_0} f(y) = L$$

It holds if $g(x) \to y_0$ for $x \to x_0$, $f(y) \to L$ for $y \to y_0$ and $g(x) \neq y_0$ in a neighborhood of x_0, except possibly at x_0; the last condition can be omitted if f is continuous at y_0. For example $\lim_{x \to 0} \frac{\sin 3x}{3x} = \lim_{y \to 0} \frac{\sin y}{y} = 1$ with $y = 3x$.

Squeeze theorem

$$g(x) \leq f(x) \leq h(x), \quad g(x) \to L, \quad h(x) \to L \implies f(x) \to L,$$

$$|f(x)| \leq C, \quad g(x) \to 0 \implies f(x)\,g(x) \to 0$$

The inequalities are required in a neighborhood of x_0, except possibly at x_0, with $L \in \mathbb{R}$ and C a constant. With an infinite limit one inequality suffices, because from $f(x) \geq g(x)$ and $g(x) \to +\infty$ it follows that $f(x) \to +\infty$. From the second line it follows, for example, that $\frac{\sin x}{x} \to 0$ for $x \to +\infty$.

Continuity and points of discontinuity

$$f \text{ continuous at } x_0 \iff \lim_{x \to x_0} f(x) = f(x_0)$$

It holds for x_0 belonging to the domain and an accumulation point of it. A point of discontinuity x_0 is called

- removable, if the limit exists finite and is different from $f(x_0)$, or $f(x_0)$ is not defined;
- a jump discontinuity (or of the first kind), if the one-sided limits exist finite and are different;
- of the second kind, if *at least one* of the one-sided limits is infinite or does not exist.

Bolzano (zeros), Weierstrass and intermediate value theorems

$$f(a)\, f(b) < 0 \implies \text{there exists } c \in (a, b) \text{ with } f(c) = 0,$$

$$\text{there exist } x_m, x_M \in [a, b] \text{ with } f(x_m) \le f(x) \le f(x_M) \text{ for every } x \in [a, b],$$

$$f\big([a, b]\big) = [m, M]$$

The three statements require f continuous on the closed and bounded interval $[a, b]$; in the third line m and M are the minimum and the maximum of f on $[a, b]$. Bolzano's theorem on zeros locates the solutions of $f(x) = 0$, the Weierstrass theorem guarantees the existence of maximum and minimum.

Vertical, horizontal and oblique asymptotes

$$x = x_0 \text{ vertical:} \quad \lim_{x \to x_0^+} f(x) = \pm\infty \ \text{ or } \ \lim_{x \to x_0^-} f(x) = \pm\infty,$$

$$y = L \text{ horizontal:} \quad \lim_{x \to +\infty} f(x) = L \in \mathbb{R},$$

$$y = mx + q \text{ oblique:} \quad \begin{cases} m = \displaystyle\lim_{x \to +\infty} \frac{f(x)}{x} \in \mathbb{R}, \ m \neq 0 \\ q = \displaystyle\lim_{x \to +\infty} \big(f(x) - mx\big) \in \mathbb{R} \end{cases}$$

For $x \to -\infty$ the definitions are analogous, and the asymptotes may be different on the two sides. The oblique asymptote exists *if and only if* both limits exist finite, with $m \neq 0$. On the same side a function has at most one horizontal or oblique asymptote.

Standard limits and hierarchy of infinities

The standard limits are proved in the chapter Limits and continuity.

Standard trigonometric limits

$$\lim_{x\to 0}\frac{\sin x}{x}=1,\qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2},\qquad \lim_{x\to 0}\frac{\tan x}{x}=1,$$

$$\lim_{x\to 0}\frac{\arcsin x}{x}=1,\qquad \lim_{x\to 0}\frac{\arctan x}{x}=1$$

They hold with x measured in radians. In place of x one can write a function $f(x)$ with $f(x)\to 0$ for $x\to x_0$ and $f(x)\neq 0$ near x_0, for example $\lim_{x\to x_0}\dfrac{\sin f(x)}{f(x)}=1$. They are used for the forms $\dfrac{0}{0}$ with trigonometric functions.

Standard limits with the number e

$$\lim_{x\to\pm\infty}\left(1+\frac{1}{x}\right)^{x}=e,\qquad \lim_{x\to\pm\infty}\left(1+\frac{a}{x}\right)^{x}=e^{a},\qquad \lim_{x\to 0}(1+x)^{\frac{1}{x}}=e$$

The second holds for every $a\in\mathbb{R}$. The number $e=2.71828\ldots$ is the base of the natural logarithms. They are used for the forms 1^{∞}.

Standard exponential and logarithmic limits

$$\lim_{x\to 0}\frac{\ln(1+x)}{x}=1,\qquad \lim_{x\to 0}\frac{e^{x}-1}{x}=1,\qquad \lim_{x\to 0}\frac{(1+x)^{\alpha}-1}{x}=\alpha,$$

$$\lim_{x\to 0}\frac{a^{x}-1}{x}=\ln a,\qquad \lim_{x\to 0}\frac{\log_a(1+x)}{x}=\frac{1}{\ln a}$$

They hold for $\alpha\in\mathbb{R}$ and $a>0$, with $a\neq 1$ in the last one. As for the trigonometric limits, in place of x one can write a function that tends to 0 and does not vanish near the point.

The form one to the infinity

$$\lim_{x\to x_0}f(x)^{g(x)}=e^{L},\qquad L=\lim_{x\to x_0}g(x)\big(f(x)-1\big)$$

It holds if $f(x)>0$, $f(x)\to 1$, $g(x)\to\pm\infty$ and the limit L exists finite; if $L=+\infty$ the limit equals $+\infty$, if $L=-\infty$ it equals 0. The formula follows from $f(x)^{g(x)}=e^{\,g(x)\ln f(x)}$ and from $\ln f(x)\sim f(x)-1$.

Asymptotic equivalences as x tends to 0

$$\sin x \sim x, \qquad \tan x \sim x, \qquad \arcsin x \sim x, \qquad \arctan x \sim x, \qquad 1 - \cos x \sim \frac{x^2}{2},$$

$$e^x - 1 \sim x, \qquad a^x - 1 \sim x \ln a, \qquad \ln(1+x) \sim x, \qquad (1+x)^\alpha - 1 \sim \alpha x$$

They hold for $x \to 0$, with $a > 0$, $a \neq 1$ and $\alpha \neq 0$, and in place of x one can write an infinitesimal function. In a product or in a quotient a factor can be replaced by an asymptotically equivalent function without changing the limit. The notation $f \sim g$ means $\frac{f(x)}{g(x)} \to 1$, the notation $f = o(g)$ means $\frac{f(x)}{g(x)} \to 0$.

Warning

Asymptotically equivalent functions are substituted in products and quotients, and substitution in sums and differences may give a wrong result. For $x \to 0$ one has $\sin x \sim x$ and $\lim_{x \to 0} \frac{\sin x - x}{x^3} = -\frac{1}{6}$, whereas substituting x for $\sin x$ would give 0. In these cases Taylor expansions are used.

Hierarchy of infinities

$$\lim_{x \to +\infty} \frac{(\ln x)^\beta}{x^\alpha} = 0, \qquad \lim_{x \to +\infty} \frac{x^\alpha}{a^x} = 0, \qquad \lim_{x \to 0^+} x^\alpha \, |\ln x|^\beta = 0$$

They hold for $\alpha > 0$, $\beta > 0$ and $a > 1$. For $x \to +\infty$ the logarithm grows more slowly than *every* power, and every power more slowly than every exponential with base greater than 1. In the forms $\frac{\infty}{\infty}$ one factors out, in the numerator and in the denominator, the term that grows fastest.

Hierarchy of infinities for sequences

$$\lim_{n \to +\infty} \frac{(\ln n)^\beta}{n^\alpha} = 0, \qquad \lim_{n \to +\infty} \frac{n^\alpha}{a^n} = 0, \qquad \lim_{n \to +\infty} \frac{a^n}{n!} = 0, \qquad \lim_{n \to +\infty} \frac{n!}{n^n} = 0$$

They hold for $\alpha > 0$, $\beta > 0$ and $a > 1$. In order of growth the infinities are $(\ln n)^\beta$, n^α, a^n, $n!$, n^n.

Standard limits of sequences

$$\lim_{n\to+\infty}\sqrt[n]{n}=1,\qquad \lim_{n\to+\infty}\sqrt[n]{a}=1,\qquad \lim_{n\to+\infty}\left(1+\frac{a}{n}\right)^n=e^a,$$

$$\lim_{n\to+\infty}q^n=\begin{cases}0 & \text{if } |q|<1\\ 1 & \text{if } q=1\\ +\infty & \text{if } q>1\end{cases}$$

The second holds for $a>0$, the third for every $a\in\mathbb{R}$. For $q\le -1$ the sequence q^n has no limit. The first limit follows from $\sqrt[n]{n}=e^{\frac{\ln n}{n}}$ and from the hierarchy of infinities.

Derivatives

The differentiation rules are proved in the chapter Differential calculus and Taylor.

Derivative, one-sided derivatives and tangent line

$$f'(x_0)=\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h},\qquad f'_{\pm}(x_0)=\lim_{h\to 0^{\pm}}\frac{f(x_0+h)-f(x_0)}{h},$$

$$y=f(x_0)+f'(x_0)\,(x-x_0)$$

The function f is differentiable at the point x_0, interior to the domain, if the first limit exists *finite*, and this happens *if and only if* the one-sided derivatives exist finite and coincide. The third formula is the tangent line to the graph at the point $(x_0, f(x_0))$. A function differentiable at x_0 is continuous at x_0.

Differentiation rules

$$(f+g)'=f'+g',\qquad (c\,f)'=c\,f',\qquad (f\,g)'=f'\,g+f\,g',$$

$$\left(\frac{f}{g}\right)'=\frac{f'\,g-f\,g'}{g^2},\qquad \left(\frac{1}{g}\right)'=-\frac{g'}{g^2}$$

They hold at the points where f and g are differentiable, with $c\in\mathbb{R}$ and $g\neq 0$ in the last two.

Derivative of the composite function and of the inverse

$$(f\circ g)'(x)=f'(g(x))\,g'(x),\qquad (f^{-1})'(y_0)=\frac{1}{f'(x_0)}\quad (y_0=f(x_0))$$

The first holds if g is differentiable at x and f is differentiable at $g(x)$; with several compositions it is applied in a chain, from the outside inward. The second holds if f is invertible and differentiable at x_0 with $f'(x_0)\neq 0$, and if f^{-1} is continuous at y_0.

Derivatives of the elementary functions

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^α	$\alpha\, x^{\alpha-1}$	$\cos x$	$-\sin x$
e^x	e^x	$\tan x$	$\dfrac{1}{\cos^2 x} = 1+\tan^2 x$
a^x	$a^x \ln a$	$\arcsin x$	$\dfrac{1}{\sqrt{1-x^2}}$
$\ln x$	$\dfrac{1}{x}$	$\arccos x$	$-\dfrac{1}{\sqrt{1-x^2}}$
$\sin x$	$\cos x$	$\arctan x$	$\dfrac{1}{1+x^2}$

Each formula holds at the interior points of the domain of the function, with $a>0$ and, for arcsin and arccos, with $-1<x<1$. The formula for x^α holds for $x>0$ and every $\alpha\in\mathbb{R}$, on all of $\mathbb{R}$ for α a positive integer and for $x\neq 0$ with α a negative integer. The derivative of a constant is 0.

Other frequently used derivatives

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
$\sqrt{x}$	$\dfrac{1}{2\sqrt{x}}$	$\log_a x$	$\dfrac{1}{x\ln a}$
$\dfrac{1}{x}$	$-\dfrac{1}{x^2}$	$\ln\lvert x\rvert$	$\dfrac{1}{x}$
$\lvert x\rvert$	$\dfrac{x}{\lvert x\rvert}$	$\sinh x$	$\cosh x$
$\sqrt[3]{x}$	$\dfrac{1}{3\sqrt[3]{x^2}}$	$\cosh x$	$\sinh x$

They hold for $x>0$ in the square root and in $\log_a x$, with $a>0$ and $a\neq 1$; for $x\neq 0$ in $\dfrac{1}{x}$, $\ln|x|$, $|x|$ and $\sqrt[3]{x}$; for every x in the hyperbolic functions. At the point 0 the functions $|x|$ and $\sqrt[3]{x}$ are not differentiable.

Derivatives of frequent composite functions

$$\left([f(x)]^\alpha\right)' = \alpha\,[f(x)]^{\alpha-1} f'(x), \qquad \left(\sqrt{f(x)}\right)' = \frac{f'(x)}{2\sqrt{f(x)}},$$

$$\left(e^{f(x)}\right)' = e^{f(x)} f'(x), \qquad \left(\ln|f(x)|\right)' = \frac{f'(x)}{f(x)},$$

$$\left(\sin f(x)\right)' = \cos f(x)\; f'(x), \qquad \left(\arctan f(x)\right)' = \frac{f'(x)}{1+[f(x)]^2}$$

They are the chain rule applied to the elementary derivatives and hold where f is differentiable and the outer function is differentiable at $f(x)$. In particular the root requires $f(x)>0$, the logarithm $f(x)\neq 0$, the power with arbitrary real α $f(x)>0$.

Derivative of a power with variable base and exponent

$$\left(f(x)^{g(x)}\right)' = f(x)^{g(x)}\left(g'(x)\ln f(x) + g(x)\,\frac{f'(x)}{f(x)}\right)$$

It holds where $f(x) > 0$ and f, g are differentiable. It is obtained by differentiating $e^{g(x)\ln f(x)}$; for example $(x^x)' = x^x(\ln x + 1)$ for $x > 0$.

Standard n-th derivatives

$$(e^{ax})^{(n)} = a^n e^{ax}, \qquad (\sin x)^{(n)} = \sin\left(x + \frac{n\pi}{2}\right), \qquad (\cos x)^{(n)} = \cos\left(x + \frac{n\pi}{2}\right),$$

$$\left(\frac{1}{1-x}\right)^{(n)} = \frac{n!}{(1-x)^{n+1}}, \qquad \big(\ln(1+x)\big)^{(n)} = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}$$

They hold for every $n \geq 1$ and $a \in \mathbb{R}$, with $x \neq 1$ in the fourth and $x > -1$ in the fifth. They give the coefficients $\dfrac{f^{(n)}(0)}{n!}$ of the Maclaurin polynomials.

Corner points, cusps and inflection points with vertical tangent

$$\text{corner point:} \quad f'_-(x_0) \neq f'_+(x_0), \text{ at least one finite,}$$

$$\text{cusp:} \quad f'_-(x_0) = \pm\infty, \quad f'_+(x_0) = \mp\infty,$$

$$\text{inflection point with vertical tangent:} \quad f'_-(x_0) = f'_+(x_0) = +\infty \text{ or } -\infty$$

The classification concerns the points where f is continuous and not differentiable, and the one-sided derivatives are the one-sided limits of the difference quotient. At the point 0 the function $|x|$ has a corner point, $\sqrt{|x|}$ a cusp, $\sqrt[3]{x}$ an inflection point with vertical tangent.

Theorems of differential calculus

The theorems with their proofs are in the chapter Differential calculus and Taylor.

Fermat's theorem

$$x_0 \text{ point of local maximum or minimum} \implies f'(x_0) = 0$$

It holds if x_0 is interior to the domain and f is differentiable at x_0. The condition $f'(x_0) = 0$ is *necessary* and not sufficient, because x^3 has zero derivative at 0 and is strictly increasing.

Rolle's theorem

$$f(a) = f(b) \implies \text{there exists } c \in (a,b) \text{ with } f'(c) = 0$$

It holds for f continuous on $[a,b]$ and differentiable on (a,b).

Mean value theorem (Lagrange)

$$\text{there exists } c \in (a,b) \text{ with } f'(c) = \frac{f(b)-f(a)}{b-a},$$

$$|f'(x)| \le K \text{ on } (a,b) \implies |f(b)-f(a)| \le K\,(b-a)$$

It holds for f continuous on $[a,b]$ and differentiable on (a,b). The estimate in the second line follows from the first and is used to prove inequalities.

Cauchy's mean value theorem

$$\text{there exists } c \in (a,b) \text{ with } \frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$$

It holds for f and g continuous on $[a,b]$ and differentiable on (a,b), with $g'(x) \neq 0$ for every $x \in (a,b)$. For $g(x) = x$ one obtains the mean value theorem.

Monotonicity and sign of the derivative

$$f' \ge 0 \text{ on } I \iff f \text{ increasing on } I,$$

$$f' \le 0 \text{ on } I \iff f \text{ decreasing on } I,$$

$$f' > 0 \text{ on } I \implies f \text{ strictly increasing on } I,$$

$$f' = 0 \text{ on } I \implies f \text{ constant on } I$$

They hold for f differentiable on an *interval* I. On a union of intervals they are applied to each interval separately. For example the function $\dfrac{1}{x}$ has negative derivative on $\mathbb{R} \setminus \{0\}$, but satisfies $f(-1) < f(1)$, therefore it is not decreasing on $\mathbb{R} \setminus \{0\}$.

Limit of the derivative

$$\lim_{x \to x_0} f'(x) = L \in \mathbb{R} \implies f \text{ differentiable at } x_0 \text{ and } f'(x_0) = L$$

It holds for f continuous at x_0 and differentiable in a neighborhood of x_0, except possibly at x_0; with the one-sided limits of f' one obtains the one-sided derivatives. It is used at the junction points of piecewise-defined functions. If the limit of f' does not exist, the criterion gives no information and one computes the limit of the difference quotient.

L'Hôpital's rule

$$\lim_{x\to x_0}\frac{f(x)}{g(x)}=\lim_{x\to x_0}\frac{f'(x)}{g'(x)}$$

It holds if the conditions

$$\begin{cases} f,g \text{ differentiable on } U\setminus\{x_0\},\ \ g'(x)\neq 0 \text{ on } U\setminus\{x_0\} \\ f,g\to 0 \ \text{ or } \ |f|,|g|\to+\infty \\ \text{the limit of } \dfrac{f'}{g'} \text{ exists, finite or infinite} \end{cases}$$

are satisfied, where U is a neighborhood of $x_0\in\mathbb{R}\cup\{\pm\infty\}$; the same holds for one-sided limits. The forms $0\cdot\infty$ and $\infty-\infty$ are reduced to a quotient, the forms 1^∞, 0^0, ∞^0 are reduced to a quotient with $f^g=e^{g\ln f}$.

Taylor and Maclaurin expansions

Taylor's formulas and the expansions of the elementary functions are derived in the chapter Differential calculus and Taylor.

Taylor's formula with Peano remainder

$$f(x)=\sum_{k=0}^{n}\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k+o\big((x-x_0)^n\big)\qquad(x\to x_0)$$

It holds if f is n times differentiable at x_0. For $x_0=0$ it is called Maclaurin's formula. The remainder is negligible with respect to the last power written, and this form is used in the computation of limits.

Lagrange remainder and error estimate

$$f(x)=\sum_{k=0}^{n}\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k+\frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)^{n+1},$$
$$\big|f(x)-T_n(x)\big|\le\frac{K}{(n+1)!}|x-x_0|^{n+1}$$

It holds if f is $n+1$ times differentiable on an interval I that contains x_0 and x, with c strictly between x_0 and x; T_n is the Taylor polynomial of order n. The estimate in the second line requires $|f^{(n+1)}|\le K$ on I.

Maclaurin expansions of exponential, sine and cosine

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots + \frac{x^n}{n!} + o(x^n),$$
$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + o(x^{2n+2}),$$
$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots + \frac{(-1)^n x^{2n}}{(2n)!} + o(x^{2n+1})$$

They hold for $x \to 0$ and every $n \in \mathbb{N}$. The sine contains *only* odd powers and the cosine only even powers, therefore the remainder is written one order higher than the last term.

Maclaurin expansions of hyperbolic sine and cosine

$$\sinh x = x + \frac{x^3}{6} + \frac{x^5}{120} + \cdots + \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2}),$$
$$\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \cdots + \frac{x^{2n}}{(2n)!} + o(x^{2n+1})$$

They hold for $x \to 0$ and every $n \in \mathbb{N}$, with the same coefficients as sine and cosine and all signs positive.

Maclaurin expansions of the logarithm and of the geometric series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + \frac{(-1)^{n+1} x^n}{n} + o(x^n),$$
$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + o(x^n),$$
$$\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-1)^n x^n + o(x^n)$$

They hold for $x \to 0$ and $n \geq 1$. With $-x$ in place of x one obtains $\ln(1-x) = -x - \frac{x^2}{2} - \cdots - \frac{x^n}{n} + o(x^n)$.

Maclaurin expansion of the binomial power

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \cdots + \binom{\alpha}{n} x^n + o(x^n),$$
$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!},$$
$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + o(x^2), \qquad \frac{1}{\sqrt{1+x}} = 1 - \frac{x}{2} + \frac{3x^2}{8} + o(x^2)$$

They hold for $x \to 0$ and every $\alpha \in \mathbb{R}$. The generalized binomial coefficient $\binom{\alpha}{n}$ coincides with the ordinary one for $\alpha \in \mathbb{N}$.

Maclaurin expansions of arctangent, tangent and arcsine

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + \frac{(-1)^n x^{2n+1}}{2n+1} + o(x^{2n+2}),$$
$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + o(x^6),$$
$$\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + o(x^6)$$

They hold for $x \to 0$. The three functions are odd, therefore their expansions contain only odd powers.

Algebra of little-o

$$o(x^n) \pm o(x^n) = o(x^n), \qquad c \cdot o(x^n) = o(x^n),$$
$$x^m \cdot o(x^n) = o(x^{m+n}), \qquad o(x^m) \cdot o(x^n) = o(x^{m+n}),$$
$$\frac{o(x^n)}{x^m} = o(x^{n-m}) \ \ (m \le n), \qquad o(x^m) = o(x^n) \ \ (m > n)$$

They hold for $x \to 0$, with m, n positive integers and $c \neq 0$, and are read from left to right. The last one lowers the order of a remainder to match it with the other terms; for the same reason a power x^m with $m > n$ is $o(x^n)$.

Expansions of composite functions and expansions centered at a point

$$f(t) = T_n(t) + o(t^n), \quad g(x) \to 0 \implies f(g(x)) = T_n(g(x)) + o\big(g(x)^n\big),$$
$$\ln x = (x-1) - \frac{(x-1)^2}{2} + o\big((x-1)^2\big) \quad (x \to 1),$$
$$e^x = e^{x_0} \sum_{k=0}^{n} \frac{(x-x_0)^k}{k!} + o\big((x-x_0)^n\big) \quad (x \to x_0)$$

In the first line $t \to 0$ and $x \to x_0$; if moreover $x_0 = 0$ and $g(x) \sim c\,x^k$ for $x \to 0$, with $c \neq 0$, the remainder $o\big(g(x)^n\big)$ is $o(x^{kn})$. The expansions centered at $x_0 \neq 0$ are obtained from the Maclaurin ones with the substitution $t = x - x_0$.

Curve sketching

The criteria for maxima, minima, convexity and inflection points are in the chapter Curve sketching.

First derivative test

$$f' > 0 \text{ to the left}, \quad f' < 0 \text{ to the right of } x_0 \implies x_0 \text{ point of local maximum},$$

$$f' < 0 \text{ to the left}, \quad f' > 0 \text{ to the right of } x_0 \implies x_0 \text{ point of local minimum}$$

It holds for f continuous at x_0 and differentiable in a neighborhood of x_0, except possibly at x_0. The test does not require differentiability at x_0, therefore it also applies to corner points and cusps. If f' has the same sign on both sides, x_0 is neither a maximum nor a minimum.

Second derivative test

$$f'(x_0) = 0, \quad f''(x_0) > 0 \implies x_0 \text{ point of local minimum},$$

$$f'(x_0) = 0, \quad f''(x_0) < 0 \implies x_0 \text{ point of local maximum}$$

It holds for f twice differentiable at x_0. For $f''(x_0) = 0$ the test is inconclusive, as shown by x^4, $-x^4$ and x^3, which at 0 have a minimum, a maximum and an inflection point.

Higher-order derivative test

$$f'(x_0) = f''(x_0) = \cdots = f^{(n-1)}(x_0) = 0, \qquad f^{(n)}(x_0) \neq 0$$

It holds for f n times differentiable at x_0, with $n \geq 2$. If n is even, x_0 is a point of local minimum for $f^{(n)}(x_0) > 0$ and of local maximum for $f^{(n)}(x_0) < 0$. If n is odd, x_0 is an inflection point with horizontal tangent.

Absolute maximum and minimum on a closed interval

$$\max_{[a,b]} f = \max\{f(a),\ f(b),\ f(x_1), \ldots, f(x_k)\},$$

$$\min_{[a,b]} f = \min\{f(a),\ f(b),\ f(x_1), \ldots, f(x_k)\}$$

It holds for f continuous on $[a, b]$, where $x_1, \ldots, x_k$ are the interior stationary points and the interior points of non-differentiability; the Weierstrass theorem guarantees the existence of maximum and minimum. On a domain that is not closed or not bounded, existence is decided with the limits at the endpoints of the domain.

Convexity and second derivative

$$f\big((1-t)\,x_1 + t\,x_2\big) \le (1-t)\,f(x_1) + t\,f(x_2),$$

$$f'' \ge 0 \text{ on } I \iff f \text{ convex on } I, \qquad f'' \le 0 \text{ on } I \iff f \text{ concave on } I,$$

$$f \text{ convex on } I \iff f(x) \ge f(x_0) + f'(x_0)\,(x - x_0)$$

The first line, required for every $x_1, x_2 \in I$ and every $t \in [0,1]$, is the definition of a convex function on the interval I. The second holds for f twice differentiable on I, and with $f'' > 0$ at the interior points the convexity is strict. The third holds for f differentiable on I, with the inequality required for every $x, x_0 \in I$, and says that the graph of a convex function lies above every tangent line.

Inflection points

$$x_0 \text{ inflection point}, \quad f \text{ twice differentiable at } x_0 \implies f''(x_0) = 0$$

The candidates are the zeros of f'' and the points where f'' does not exist. A candidate at which f is continuous and the graph has a tangent line is an inflection point if f'' changes sign at the point. The tangent at the inflection point is $y = f(x_0) + f'(x_0)(x - x_0)$, and the graph crosses it.

Basic antiderivatives

The elementary antiderivatives are derived in the chapter Integral calculus.

Antiderivative and linearity

$$\int f(x)\,dx = F(x) + c \iff F'(x) = f(x),$$

$$\int \big(\alpha f(x) + \beta g(x)\big)\,dx = \alpha \int f(x)\,dx + \beta \int g(x)\,dx$$

They hold on an interval I, with $\alpha, \beta \in \mathbb{R}$ and $c \in \mathbb{R}$ an arbitrary constant, because two antiderivatives of the same function on an interval differ by a constant. Every function continuous on an interval has antiderivatives.

Elementary integrals

$$\int x^{\alpha}\,dx = \frac{x^{\alpha+1}}{\alpha+1} + c \ \ (\alpha \neq -1), \qquad \int \frac{1}{x}\,dx = \ln|x| + c,$$

$$\int e^{x}\,dx = e^{x} + c, \qquad \int a^{x}\,dx = \frac{a^{x}}{\ln a} + c \ \ (a > 0,\ a \neq 1),$$

$$\int \sin x\,dx = -\cos x + c, \qquad \int \cos x\,dx = \sin x + c$$

Each holds on every interval contained in the domain of the integrand; for $\frac{1}{x}$ the intervals are $(-\infty, 0)$ and $(0, +\infty)$. Frequent cases of the first are $\int \sqrt{x}\,dx = \frac{2}{3}\,x^{\frac{3}{2}} + c$ and $\int \frac{1}{x^2}\,dx = -\frac{1}{x} + c$.

Elementary trigonometric and hyperbolic integrals

$$\int \frac{1}{\cos^2 x}\,dx = \tan x + c, \qquad \int \frac{1}{\sin^2 x}\,dx = -\frac{\cos x}{\sin x} + c,$$

$$\int \frac{1}{1+x^2}\,dx = \arctan x + c, \qquad \int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + c,$$

$$\int \sinh x\,dx = \cosh x + c, \qquad \int \cosh x\,dx = \sinh x + c$$

They hold on the intervals where the integrand is defined, that is with $\cos x \neq 0$ in the first, $\sin x \neq 0$ in the second and $-1 < x < 1$ in the fourth.

Integrals with linear argument

$$\int f(ax+b)\,dx = \frac{1}{a}\,F(ax+b) + c,$$

$$\int \frac{dx}{ax+b} = \frac{1}{a}\ln|ax+b| + c, \qquad \int e^{ax}\,dx = \frac{e^{ax}}{a} + c,$$

$$\int \sin(ax)\,dx = -\frac{\cos(ax)}{a} + c, \qquad \int \cos(ax)\,dx = \frac{\sin(ax)}{a} + c$$

They hold for $a \neq 0$, with F an antiderivative of f, and follow from the substitution $t = ax + b$.

Near-elementary integrals with power, logarithm and exponential

$$\int f'(x)\,[f(x)]^{\alpha}\,dx = \frac{[f(x)]^{\alpha+1}}{\alpha+1} + c \quad (\alpha \neq -1),$$

$$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c, \qquad \int f'(x)\,e^{f(x)}\,dx = e^{f(x)} + c,$$

$$\int \frac{f'(x)}{\sqrt{f(x)}}\,dx = 2\sqrt{f(x)} + c$$

They hold for f differentiable with continuous derivative, on the intervals where the integrand is defined. The integrand contains the derivative of the inner function, possibly up to a multiplicative constant, as in $\int x\,e^{x^2}\,dx = \frac{1}{2}\,e^{x^2} + c$.

Near-elementary integrals with trigonometric and inverse trigonometric functions

$$\int f'(x)\cos f(x)\,dx = \sin f(x) + c, \qquad \int f'(x)\sin f(x)\,dx = -\cos f(x) + c,$$

$$\int \frac{f'(x)}{\cos^2 f(x)}\,dx = \tan f(x) + c, \qquad \int \frac{f'(x)}{1+[f(x)]^2}\,dx = \arctan f(x) + c,$$

$$\int \frac{f'(x)}{\sqrt{1-[f(x)]^2}}\,dx = \arcsin f(x) + c$$

They hold for f differentiable with continuous derivative, on the intervals where the integrand is defined. For example $\int \frac{e^x}{1+e^{2x}}\,dx = \arctan e^x + c$.

Integrals with a positive constant a

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a} + c, \qquad \int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + c,$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + c, \qquad \int \frac{dx}{\sqrt{x^2+k}} = \ln\left|x+\sqrt{x^2+k}\right| + c$$

The first three hold for $a > 0$, the second for $|x| < a$, the third for $x \neq \pm a$. The last one holds for $k \neq 0$ on the intervals where $x^2 + k > 0$.

Other frequently used antiderivatives

$$\int \tan x\,dx = -\ln|\cos x| + c, \qquad \int \ln x\,dx = x\ln x - x + c,$$

$$\int \sin^2 x\,dx = \frac{x - \sin x\cos x}{2} + c, \qquad \int \cos^2 x\,dx = \frac{x + \sin x\cos x}{2} + c,$$

$$\int \arctan x\,dx = x\arctan x - \frac{1}{2}\ln(1+x^2) + c,$$

$$\int \sqrt{1-x^2}\,dx = \frac{\arcsin x + x\sqrt{1-x^2}}{2} + c$$

They hold on the intervals where the integrand is defined, that is with $\cos x \neq 0$ in the first, $x > 0$ in the second and $-1 \leq x \leq 1$ in the last. They are obtained by parts, by substitution or with the power-reduction formulas.

Warning

The integral of a product is different from the product of the integrals. One has $\int x \cdot x\,dx = \frac{x^3}{3} + c$, whereas the product of the antiderivatives $\frac{x^2}{2} \cdot \frac{x^2}{2} = \frac{x^4}{4}$ has derivative x^3. For products one uses integration by parts or substitution.

Integration methods

The methods are proved, with examples, in the chapter Integral calculus.

Integration by parts

$$\int f(x)\,g'(x)\,dx = f(x)\,g(x) - \int f'(x)\,g(x)\,dx,$$

$$\int_a^b f(x)\,g'(x)\,dx = \big[f(x)\,g(x)\big]_a^b - \int_a^b f'(x)\,g(x)\,dx$$

It holds for f and g differentiable with continuous derivative on an interval. In products of a polynomial by e^x, $\sin x$ or $\cos x$ the factor f is the polynomial; with $\ln x$ and $\arctan x$ one takes $f(x) = \ln x$ or $f(x) = \arctan x$ and $g'(x) = 1$.

Exponential times sine and cosine

$$\int e^{ax}\sin(bx)\,dx = \frac{e^{ax}\big(a\sin(bx) - b\cos(bx)\big)}{a^2+b^2} + c,$$

$$\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}\big(a\cos(bx) + b\sin(bx)\big)}{a^2+b^2} + c$$

They hold for $a, b \in \mathbb{R}$ not both zero. They are obtained by integrating by parts twice; at the second step the original integral reappears, and the equality obtained is solved for it.

Integration by substitution

$$\int f(g(x))\,g'(x)\,dx = F(g(x)) + c \qquad \big(t = g(x),\ dt = g'(x)\,dx\big),$$

$$\int f(x)\,dx = \int f(\varphi(t))\,\varphi'(t)\,dt \qquad \big(x = \varphi(t),\ dx = \varphi'(t)\,dt\big),$$

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(t)\,dt$$

They hold for f continuous, with antiderivative F, and for g, φ differentiable with continuous derivative. In the inverse substitution φ is invertible, and once the antiderivative in t is computed one returns to the variable x with $t = \varphi^{-1}(x)$. In the definite integral the limits of integration are transformed with the substitution and the computation is completed in the variable t.

Typical substitutions

function of e^x	$t = e^x, \quad dx = \dfrac{dt}{t}$
$\ln x$ with the factor $\dfrac{1}{x}$	$t = \ln x, \quad dt = \dfrac{dx}{x}$
$\sqrt[n]{ax+b}$	$t = \sqrt[n]{ax+b}, \quad dx = \dfrac{n\,t^{n-1}}{a}\,dt$
$\sqrt{a^2-x^2}$	$x = a\sin t, \quad dx = a\cos t\,dt$
rational in $\sin x,\ \cos x$	$t = \tan\dfrac{x}{2}, \quad dx = \dfrac{2}{1+t^2}\,dt$

In the third line $a \neq 0$. In the fourth $a > 0$ and $t \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$, so that $\sqrt{a^2-x^2} = a\cos t$. In the last one $x \in (-\pi, \pi)$, and sine and cosine are expressed with the half-angle substitution formulas.

Partial fraction decomposition

factor of $Q(x)$	terms of the decomposition
$x - x_0$	$\dfrac{A}{x - x_0}$
$(x - x_0)^m$	$\dfrac{A_1}{x - x_0} + \dfrac{A_2}{(x - x_0)^2} + \cdots + \dfrac{A_m}{(x - x_0)^m}$
$x^2 + px + q,\ \ p^2 - 4q < 0$	$\dfrac{Ax + B}{x^2 + px + q}$

It holds for $\dfrac{P(x)}{Q(x)}$ with the degree of P less than the degree of Q; otherwise one first performs the division with remainder, $\dfrac{P}{Q} = S + \dfrac{R}{Q}$. The constants are determined by the principle of identity of polynomials or by substituting the roots of Q.

Integrals of partial fractions

$$\int \frac{A}{x - x_0}\,dx = A \ln|x - x_0| + c, \qquad \int \frac{A}{(x - x_0)^k}\,dx = -\frac{A}{(k-1)(x - x_0)^{k-1}} + c \ \ (k \geq 2),$$

$$\int \frac{Ax + B}{x^2 + px + q}\,dx = \frac{A}{2}\ln(x^2 + px + q) + \left(B - \frac{Ap}{2}\right)\int \frac{dx}{x^2 + px + q}$$

The third holds for $p^2 - 4q < 0$ and is obtained by writing $Ax + B = \dfrac{A}{2}(2x + p) + B - \dfrac{Ap}{2}$, so that the first term contains the derivative of the denominator.

Integral of the reciprocal of a quadratic

$$\int \frac{dx}{x^2 + px + q} = \begin{cases} \dfrac{2}{\sqrt{-\Delta}} \arctan \dfrac{2x + p}{\sqrt{-\Delta}} + c & \text{if } \Delta < 0 \\ -\dfrac{2}{2x + p} + c & \text{if } \Delta = 0 \\ \dfrac{1}{\sqrt{\Delta}} \ln \left| \dfrac{2x + p - \sqrt{\Delta}}{2x + p + \sqrt{\Delta}} \right| + c & \text{if } \Delta > 0 \end{cases}$$

Here $\Delta = p^2 - 4q$. The case $\Delta < 0$ is obtained by completing the square, $x^2 + px + q = \left(x + \dfrac{p}{2}\right)^2 + \dfrac{-\Delta}{4}$, and the case $\Delta > 0$ with partial fractions.

Definite integral and areas

The fundamental theorem and the properties of the integral are in the chapter Integral calculus.

Properties of the definite integral

$$\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx, \qquad \int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx,$$

$$\int_b^a f\,dx = -\int_a^b f\,dx, \qquad f \le g \implies \int_a^b f\,dx \le \int_a^b g\,dx, \qquad \left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx$$

They hold for f and g integrable, with $\alpha, \beta \in \mathbb{R}$; monotonicity and the absolute value inequality require $a < b$. Additivity is used for piecewise-defined functions and for those with the absolute value.

Integrals of even, odd and periodic functions

$$\int_{-a}^a f(x)\,dx = 0 \ \ (f \text{ odd}), \qquad \int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx \ \ (f \text{ even}),$$

$$\int_c^{c+T} f(x)\,dx = \int_0^T f(x)\,dx \ \ (f \text{ periodic with period } T)$$

The first two hold for f integrable on $[-a, a]$ with $a > 0$, the third for f integrable on a period and every $c \in \mathbb{R}$.

Mean value theorem for integrals

$$\frac{1}{b-a}\int_a^b f(x)\,dx = f(c) \quad \text{for some } c \in [a, b]$$

It holds for f continuous on $[a, b]$, with $a < b$. The left-hand side is the integral mean of f on $[a, b]$.

Fundamental theorem of calculus

$$F(x) = \int_a^x f(t)\,dt \implies F'(x) = f(x), \qquad \int_a^b f(x)\,dx = G(b) - G(a) = \big[G(x)\big]_a^b$$

They hold for f continuous on $[a, b]$, and in the second G is any antiderivative of f. The integral function F is therefore an antiderivative of f, with $F(a) = 0$.

Derivative of an integral function with variable limits

$$\frac{d}{dx}\int_{\alpha(x)}^{\beta(x)} f(t)\,dt = f(\beta(x))\,\beta'(x) - f(\alpha(x))\,\alpha'(x)$$

It holds for α and β differentiable and f continuous on an interval that contains the values of α and β. For example $\dfrac{d}{dx}\displaystyle\int_0^{x^3} \cos(t^2)\,dt = 3x^2\cos(x^6)$.

Area between two curves

$$\mathcal{A} = \int_a^b \big(f(x) - g(x)\big)\,dx, \qquad \mathcal{A} = \int_a^b |f(x)|\,dx$$

The first is the area of the region between the graphs of f and g and holds for f, g continuous on $[a, b]$ with $f(x) \geq g(x)$; the second is the area between the graph of f and the x-axis. If the graphs intersect, the interval is split at the points where $f(x) = g(x)$ and on each piece one integrates the difference between the larger and the smaller function.

Improper integrals

The definitions and the convergence tests are in the chapter Improper integrals.

Definition of improper integral

$$\int_a^{+\infty} f(x)\,dx = \lim_{t \to +\infty} \int_a^t f(x)\,dx, \qquad \int_a^b f(x)\,dx = \lim_{t \to a^+} \int_t^b f(x)\,dx,$$

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_c^{+\infty} f(x)\,dx$$

The first holds for f integrable on every $[a, t]$; the second for f unbounded near a and integrable on every $[t, b]$ with $a < t < b$. The integral converges if the limit exists finite, diverges if it is infinite, is indeterminate if the limit does not exist. An infinite endpoint, or a point near which f is unbounded, is called a **critical point** of the integral. With several critical points the integral is split into integrals with a single critical point and converges if *all* of them converge; the behavior does not depend on c.

Reference integrals of powers

$$\int_1^{+\infty} \frac{dx}{x^\alpha} = \frac{1}{\alpha - 1} \quad (\alpha > 1), \qquad \int_0^1 \frac{dx}{x^\alpha} = \frac{1}{1 - \alpha} \quad (\alpha < 1)$$

The first integral converges *if and only if* $\alpha > 1$, the second *if and only if* $\alpha < 1$, and in the other cases they diverge to $+\infty$. Therefore $\displaystyle\int_0^{+\infty} \frac{dx}{x^\alpha}$ diverges for every $\alpha \in \mathbb{R}$.

Reference integrals with the critical point at an arbitrary endpoint

$$\int_a^b \frac{dx}{(x-a)^\alpha} \text{ and } \int_a^b \frac{dx}{(b-x)^\alpha} \text{ converge } \iff \alpha < 1,$$

$$\int_c^{+\infty} \frac{dx}{x^\alpha} \text{ converges } \iff \alpha > 1$$

They hold for $a < b$ and $c > 0$. Moving the endpoint away from the critical point does not change the behavior.

Reference integrals with exponential and logarithm

$$\int_0^{+\infty} e^{-\lambda x}\,dx = \frac{1}{\lambda} \ (\lambda > 0), \qquad \int_2^{+\infty} \frac{dx}{x\,(\ln x)^\beta} \text{ converges } \iff \beta > 1,$$

$$\int_0^{\frac{1}{2}} \frac{dx}{x\,|\ln x|^\beta} \text{ converges } \iff \beta > 1$$

For $\lambda \leq 0$ the first integral diverges. The reference integrals with the logarithm are reduced to those of powers with the substitution $u = \ln x$, and are needed when comparison with powers is inconclusive.

Comparison test for integrals

$$0 \leq f(x) \leq g(x), \quad \int g \text{ converges } \implies \int f \text{ converges},$$

$$0 \leq f(x) \leq g(x), \quad \int f \text{ diverges } \implies \int g \text{ diverges}$$

It holds with the inequalities satisfied near the critical point, for f and g integrable on every closed and bounded interval that does not contain the critical point. The integral of a nonnegative function either converges or diverges to $+\infty$.

Limit comparison test for integrals

$$f(x) \sim g(x) \ (x \to c) \implies \int f \text{ and } \int g \text{ have the same behavior}$$

It holds for f and g positive near the critical point c, finite or infinite; in practice g is a reference integrand, obtained with the standard limits or with Taylor expansions. More generally, setting $L = \lim_{x \to c} \frac{f(x)}{g(x)}$,

- if $0 < L < +\infty$, the integrals of f and of g have the same behavior;
- if $L = 0$ and the integral of g converges, the integral of f converges as well;
- if $L = +\infty$ and the integral of g diverges, the integral of f diverges as well.

Order of infinitesimal and of infinity of the integrand

$$f(x) \sim \frac{k}{x^p} \ (x \to +\infty), \qquad f(x) \sim \frac{k}{|x - x_0|^p} \ (x \to x_0)$$

It holds for f positive, $k > 0$ and $p > 0$. For $x \to +\infty$ the integral converges if $p > 1$ and diverges if $p \leq 1$; at the finite critical point x_0 the integral converges if $p < 1$ and diverges if $p \geq 1$.

Absolute convergence of integrals

$$\int_a^{+\infty} |f(x)|\,dx \text{ converges} \implies \int_a^{+\infty} f(x)\,dx \text{ converges}, \qquad \left|\int_a^{+\infty} f(x)\,dx\right| \leq \int_a^{+\infty} |f(x)|\,dx$$

It holds for f integrable on every closed and bounded interval contained in $[a, +\infty)$, and the same statement holds for finite critical points. It is used for integrands of variable sign, for example with $\left|\frac{\cos x}{x^2}\right| \leq \frac{1}{x^2}$ on $[1, +\infty)$. The converse implication is false, as shown by $\int_1^{+\infty} \frac{\sin x}{x}\,dx$, which converges while the integral of the absolute value diverges.

Standard improper integrals

$$\int_0^{+\infty} e^{-x}\,dx = 1, \qquad \int_{-\infty}^{+\infty} e^{-x^2}\,dx = \sqrt{\pi}, \qquad \int_0^{+\infty} \frac{\sin x}{x}\,dx = \frac{\pi}{2}$$

The third integral converges and does not converge absolutely. At 0 the function $\frac{\sin x}{x}$ tends to 1, therefore 0 is not a critical point.

Warning

The fundamental formula of integral calculus applies to a function continuous on the whole closed interval of integration. The function $\frac{1}{x^2}$ is not defined at 0, and $-\frac{1}{x}$ is an antiderivative of it only on the intervals that do not contain 0. Therefore the formula does not apply to $\int_{-1}^{1} \frac{dx}{x^2}$, and the computation $\left[-\frac{1}{x}\right]_{-1}^{1} = -2$ would give a negative value for a positive function. The point 0 is a critical point interior to the interval, therefore the integral is split into the integrals on $[-1, 0)$ and on $(0, 1]$; since both diverge to $+\infty$, the original integral diverges as well.

Numerical series and power series

The tests and the reference series are proved in the chapter Numerical series.

Partial sums and necessary condition

$$s_N = \sum_{n=1}^{N} a_n, \qquad \sum_{n=1}^{\infty} a_n = \lim_{N\to+\infty} s_N, \qquad \sum_{n=1}^{\infty} a_n \text{ converges} \implies \lim_{n\to+\infty} a_n = 0$$

The series converges if the limit of the partial sums exists finite, diverges if it is infinite, is indeterminate if it does not exist. The condition $a_n \to 0$ is *necessary* and not sufficient, as shown by the harmonic series. The behavior does not change if finitely many terms are modified.

Geometric series

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}, \qquad \sum_{n=n_0}^{\infty} q^n = \frac{q^{n_0}}{1-q} \qquad (|q|<1)$$

The series converges *if and only if* $|q|<1$, diverges to $+\infty$ for $q \geq 1$ and is indeterminate for $q \leq -1$. The sum is the first term divided by $1-q$.

Telescoping series and Mengoli's series

$$\sum_{n=1}^{\infty} (b_n - b_{n+1}) = b_1 - \lim_{n\to+\infty} b_n, \qquad \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1$$

The telescoping series converges *if and only if* the sequence (b_n) converges, and in that case the first formula holds. The form $a_n = b_n - b_{n+1}$ is usually obtained with partial fractions.

Generalized harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}} \text{ converges} \iff \alpha > 1,$$

$$\sum_{n=2}^{\infty} \frac{1}{n^{\alpha}(\ln n)^{\beta}} \text{ converges} \iff \alpha > 1 \text{ or } \alpha = 1,\ \beta > 1$$

They hold for $\alpha, \beta \in \mathbb{R}$, and when they do not converge the two series diverge to $+\infty$. For $\alpha = 1$ the first is the harmonic series, which diverges. They are the reference terms of the limit comparison test.

Comparison and limit comparison tests

$$0 \le a_n \le b_n, \quad \sum b_n \text{ converges} \implies \sum a_n \text{ converges},$$

$$0 \le a_n \le b_n, \quad \sum a_n \text{ diverges} \implies \sum b_n \text{ diverges},$$

$$a_n \sim b_n \implies \sum a_n \text{ and } \sum b_n \text{ have the same behavior}$$

They hold for series with eventually nonnegative terms, with $b_n > 0$ in the third line. More generally, if $\dfrac{a_n}{b_n} \to \ell \in (0, +\infty)$ the two series have the same behavior; if $\ell = 0$ the convergence of $\sum b_n$ implies that of $\sum a_n$; if $\ell = +\infty$ the divergence of $\sum b_n$ implies that of $\sum a_n$.

Ratio test and root test

$$\ell = \lim_{n \to +\infty} \frac{a_{n+1}}{a_n} \text{ or } \ell = \lim_{n \to +\infty} \sqrt[n]{a_n}, \qquad \ell < 1 \implies \sum a_n \text{ converges}, \qquad \ell > 1 \implies \sum a_n \text{ diver}$$

The ratio test holds for $a_n > 0$, the root test for $a_n \ge 0$ eventually, with ℓ finite or $+\infty$. For $\ell = 1$ the tests give no information, because $\sum \dfrac{1}{n}$ and $\sum \dfrac{1}{n^2}$ both give $\ell = 1$. The ratio test is suited to factorials, the root test to n-th powers.

Integral test

$$\sum_{n=n_0}^{\infty} f(n) \text{ converges} \iff \int_{n_0}^{+\infty} f(x)\,dx \text{ converges}$$

It holds for $f : [n_0, +\infty) \to \mathbb{R}$ nonnegative and decreasing. With $f(x) = \dfrac{1}{x^\alpha}$ it gives the behavior of the generalized harmonic series.

Leibniz test

$$\sum_{n=1}^{\infty} (-1)^n b_n \text{ converges}, \qquad |s - s_N| \le b_{N+1}$$

It holds if the conditions

$$\begin{cases} b_n \ge 0 \\ b_{n+1} \le b_n \\ \lim\limits_{n \to +\infty} b_n = 0 \end{cases}$$

are satisfied for every n, where s is the sum of the series and s_N the partial sum. The error made by stopping at s_N does not exceed the first neglected term.

Absolute convergence of series

$$\sum_{n=1}^{\infty} |a_n| \text{ converges} \implies \sum_{n=1}^{\infty} a_n \text{ converges}, \qquad \left|\sum_{n=1}^{\infty} a_n\right| \le \sum_{n=1}^{\infty} |a_n|$$

It reduces a series of arbitrary sign to a series with nonnegative terms. The converse implication is false, because $\sum \dfrac{(-1)^n}{n}$ converges by the Leibniz test, whereas the series of the absolute values is the harmonic series.

Sums of standard series and linearity

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e, \qquad \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \ln 2, \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

$$\sum_{n=1}^{\infty} (\alpha\, a_n + \beta\, b_n) = \alpha \sum_{n=1}^{\infty} a_n + \beta \sum_{n=1}^{\infty} b_n$$

Linearity holds if $\sum a_n$ and $\sum b_n$ converge, with $\alpha, \beta \in \mathbb{R}$. The first two sums follow from the Taylor series of e^x and of $\ln(1+x)$ with $x = 1$; the third is taken as known.

Power series and radius of convergence

$$\sum_{n=0}^{\infty} c_n (x - x_0)^n, \qquad R = \frac{1}{\ell}, \qquad \ell = \lim_{n \to +\infty} \frac{|c_{n+1}|}{|c_n|} \text{ or } \ell = \lim_{n \to +\infty} \sqrt[n]{|c_n|}$$

The series converges absolutely for $|x - x_0| < R$ and does not converge for $|x - x_0| > R$; at the endpoints $x_0 - R$ and $x_0 + R$ the behavior is studied by substituting the value. The formula for the radius holds if the limit ℓ exists, finite or $+\infty$, with $R = +\infty$ for $\ell = 0$ and $R = 0$ for $\ell = +\infty$; the limit of the ratio requires $c_n \neq 0$ eventually.

Elementary Taylor series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \qquad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \qquad \ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}, \qquad \arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$$

The first three hold for every $x \in \mathbb{R}$; the geometric series for $-1 < x < 1$, that of the logarithm for $-1 < x \le 1$, that of the arctangent for $-1 \le x \le 1$.

Differential equations

The solution methods are proved in the chapter Differential equations.

Cauchy problem, existence and uniqueness

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases}$$

If f and $\dfrac{\partial f}{\partial y}$ are continuous on an open rectangle that contains (x_0, y_0), the problem has one and only one solution, defined in a neighborhood of x_0. The constant of the general solution is obtained by substituting x_0 and y_0.

Separable differential equations

$$y' = g(x)\, h(y) \implies \int \frac{dy}{h(y)} = \int g(x)\, dx + C$$

It holds in the regions where $h(y) \neq 0$, with g continuous and h of class C^1. Every $\bar{y}$ with $h(\bar{y}) = 0$ gives the constant solution $y \equiv \bar{y}$, which is added separately to the general solution.

First-order linear equations

$$y' + a(x)\, y = b(x) \implies y(x) = e^{-A(x)} \left(C + \int b(x)\, e^{A(x)}\, dx \right),$$

$$y(x) = e^{-A(x)} \left(y_0 + \int_{x_0}^{x} b(t)\, e^{A(t)}\, dt \right)$$

They hold for a and b continuous on an interval I, with A an antiderivative of a and $C \in \mathbb{R}$. The second line is the solution of the Cauchy problem with $y(x_0) = y_0$, in which $A(x) = \displaystyle\int_{x_0}^{x} a(t)\, dt$ is the antiderivative of a that vanishes at x_0. The solutions are defined on all of I and have the form $Ce^{-A(x)} + y_p(x)$, with y_p a particular solution.

Bernoulli equation

$$y' + a(x)\, y = b(x)\, y^{\alpha}, \qquad z = y^{1-\alpha} \implies z' + (1-\alpha)\, a(x)\, z = (1-\alpha)\, b(x)$$

It holds for $\alpha \neq 0$, $\alpha \neq 1$, in the regions where $y \neq 0$, with $y > 0$ if α is not an integer. For $\alpha > 0$ also $y \equiv 0$ is a solution.

Second-order homogeneous equations with constant coefficients

$$y'' + ay' + by = 0, \qquad r^2 + ar + b = 0, \qquad \Delta = a^2 - 4b$$

$$y(x) = \begin{cases} C_1 e^{r_1 x} + C_2 e^{r_2 x} & \text{if } \Delta > 0 \\ (C_1 + C_2 x)\, e^{rx} & \text{if } \Delta = 0 \\ e^{\lambda x}\big(C_1 \cos(\mu x) + C_2 \sin(\mu x)\big) & \text{if } \Delta < 0 \end{cases}$$

They hold for $a, b \in \mathbb{R}$ and $C_1, C_2 \in \mathbb{R}$. Here $r_1 \neq r_2$ are the real roots, $r = -\dfrac{a}{2}$ the double root, $\lambda \pm i\mu$ the complex conjugate roots, with $\lambda = -\dfrac{a}{2}$ and $\mu = \dfrac{\sqrt{-\Delta}}{2}$.

Structure of the general solution

$$y = C_1 y_1 + C_2 y_2 + y_p$$

It holds for $y'' + ay' + by = f(x)$ with f continuous, where y_1 and y_2 are non-proportional solutions of the associated homogeneous equation and y_p is any particular solution of the complete equation. If $f = f_1 + f_2$, the sum of the particular solutions relative to f_1 and to f_2 is a particular solution relative to f.

Method of undetermined coefficients

$f(x)$	trial y_p	associated number
$p_n(x)$	$q_n(x)$	0
$A\, e^{\gamma x}$	$a_0\, e^{\gamma x}$	γ
$p_n(x)\, e^{\gamma x}$	$q_n(x)\, e^{\gamma x}$	γ
$A\cos(\omega x) + B\sin(\omega x)$	$a_0\cos(\omega x) + b_0\sin(\omega x)$	$\pm i\omega$
$e^{\gamma x}\big(A\cos(\omega x) + B\sin(\omega x)\big)$	$e^{\gamma x}\big(a_0\cos(\omega x) + b_0\sin(\omega x)\big)$	$\gamma \pm i\omega$

Here p_n is a polynomial of degree n and q_n a polynomial of degree n with coefficients to be determined. If the associated number is a simple root of the characteristic equation, the trial solution is multiplied by x; if it is a double root, by x^2. The coefficients are found by substituting y_p into the equation.

Variation of parameters

$$y_p = -y_1 \int \frac{y_2\, f}{W}\, dx + y_2 \int \frac{y_1\, f}{W}\, dx, \qquad W = y_1 y_2' - y_1' y_2$$

It holds for f continuous, with y_1 and y_2 non-proportional solutions of the homogeneous equation associated with $y'' + ay' + by = f(x)$; the Wronskian W never vanishes. It also applies to forcing terms that do not appear in the table of the method of undetermined coefficients.

Second-order Cauchy problem

$$\begin{cases} y'' + ay' + by = f(x) \\ y(x_0) = y_0 \\ y'(x_0) = v_0 \end{cases}$$

With f continuous on an interval I and $x_0 \in I$ the problem has one and only one solution, defined on all of I. The constants C_1 and C_2 are obtained by imposing the two conditions on the general solution and on its derivative.

Linear homogeneous equations of order n

$$y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0,$$

$$r^n + a_{n-1}r^{n-1} + \cdots + a_1 r + a_0 = 0$$

It holds for real constant coefficients $a_0, \ldots, a_{n-1}$. A real root r of multiplicity m gives the solutions $e^{rx}, xe^{rx}, \ldots, x^{m-1}e^{rx}$; a pair of roots $\lambda \pm i\mu$ of multiplicity m gives $x^k e^{\lambda x}\cos(\mu x)$ and $x^k e^{\lambda x}\sin(\mu x)$ for $k = 0, \ldots, m-1$. The general solution is the linear combination of the n solutions thus obtained.